

Cloud Security Monitoring Investigation Report

Project Title

GCP Security Monitoring and Logging Lab

Analyst: Nitin Sukthe

Role: Cloud Security Engineer (Project Simulation)

Platform: Google Cloud Platform (GCP)

Environment: Personal GCP Project

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Section 1: Identify the Cloud Security Services Involved

The primary cloud security services involved in this investigation were Google Cloud Audit Logs, Cloud Logging, Log-Based Metrics, Cloud Monitoring, Identity and Access Management (IAM), and Compute Engine.

Cloud Audit Logs were used to capture administrative actions and service activities occurring within the Google Cloud environment. Cloud Logging was used as the centralized platform for collecting and reviewing security-relevant events generated by cloud resources and APIs. Log-Based Metrics were implemented to transform audit log entries into measurable security events, enabling automated monitoring and alert generation.

Cloud Monitoring was configured to provide real-time visibility into cloud activity through alert policies. Compute Engine was deployed as a monitored resource to generate cloud activity and validate monitoring controls. IAM was reviewed to identify access control risks and evaluate adherence to the Principle of Least Privilege.

The objective of using these services was to establish visibility into cloud operations, validate logging functionality, implement detection capabilities, and assess the security posture of the cloud environment.

Section 2: Document the Investigation

A cloud security monitoring project was created within Google Cloud Platform to evaluate native monitoring and logging capabilities. The project began with the creation of a dedicated cloud environment followed by the enablement of security-related APIs, including Cloud Logging, Cloud Monitoring, Compute Engine, Cloud Asset API, and Security Command Center API.

The analyst attempted to access Security Command Center to assess cloud security posture. During testing, it was determined that Security Command Center required an organizational account and was unavailable within the personal GCP environment. As a result, alternative security monitoring controls were implemented using Cloud Logging, Cloud Audit Logs, Log-Based Metrics, and Cloud Monitoring.

The analyst then deployed a Compute Engine virtual machine to generate monitored cloud activity. After resource creation, Cloud Logging was used to investigate audit events generated by administrative actions and API activity. Queries were executed within Logs Explorer to identify security-relevant records associated with project operations.

During log analysis, Cloud Audit Logs recorded multiple administrative events, including service enablement operations, resource management activities, Cloud OS Config events, and service usage actions. These logs provided visibility into cloud administrative behavior and demonstrated successful audit log collection.

To enhance monitoring capabilities, the analyst created a custom Log-Based Metric named "vm_activity_metric." This metric was configured to detect Compute Engine-related activity and convert audit log events into measurable monitoring data. The metric was validated through additional cloud activity generated by Compute Engine operations.

Following successful metric creation, a Cloud Monitoring Alert Policy named "GCE VM Activity Alert" was configured. The policy monitored activity generated by the custom metric and provided automated detection capabilities whenever monitored activity exceeded defined thresholds. This configuration established an end-to-end monitoring workflow from cloud activity generation through alert creation.

The final phase of the investigation focused on Identity and Access Management (IAM). During the review, the analyst observed that the project account possessed the Owner role, providing unrestricted administrative access. Additionally, the default Compute Engine service account was assigned the Editor role, granting broad permissions across multiple services. While these permissions are commonly observed in newly created projects, they represent security risks that should be addressed through least-privilege access controls.

Based on the investigation, the monitoring environment successfully collected audit logs, generated security telemetry, implemented automated alerting, and identified IAM-related security improvement opportunities.

Section 3: Security Findings

Finding F-01

Title

Owner Role Assigned to Project User

Severity

Medium

Observation

The project user account possessed the Owner role.

Risk

Owner permissions provide unrestricted administrative access to cloud resources, IAM settings, billing configurations, and security controls.

Impact

Compromise of the Owner account could result in unauthorized modifications, resource deletion, privilege escalation, and loss of security governance.

Finding F-02**Title**

Default Service Account Assigned Editor Permissions

Severity

High

Observation

The Compute Engine default service account possessed the Editor role.

Risk

Editor permissions grant broad access to numerous cloud services and resources.

Impact

Compromise of a workload utilizing this service account could result in excessive cloud access and potential lateral movement within the environment.

Finding F-03

Title

Security Command Center Unavailable

Severity

Informational

Observation

Security Command Center required an organization-level environment and was unavailable in the personal project.

Impact

Advanced posture management and threat detection capabilities could not be evaluated within the test environment.

Mitigation

Cloud Logging, Cloud Audit Logs, Log-Based Metrics, and Cloud Monitoring were successfully used as alternative monitoring controls.

Section 4: Security Recommendations

Recommendation 1: Apply Least Privilege Access Controls

Administrative accounts should be assigned only the permissions necessary to perform their job responsibilities. Broad roles such as Owner should be restricted whenever possible.

Recommendation 2: Harden Service Account Permissions

Default service accounts should not retain Editor permissions. Dedicated workload identities should be created with narrowly scoped IAM permissions.

Recommendation 3: Expand Monitoring Coverage

Additional monitoring rules should be implemented for:

- IAM policy modifications
- Service account activity

- Resource deletion events
- Billing changes
- Security configuration changes

Recommendation 4: Centralize Security Logging

Cloud logs should be exported to a centralized SIEM platform for long-term retention, correlation, and incident investigation.

Recommendation 5: Enable Organization-Level Security Services

In enterprise environments, Security Command Center should be enabled to provide posture management, vulnerability assessment, asset visibility, and threat detection capabilities.

Risk Assessment Summary

Finding	Severity	Risk Level
Owner Role Assignment	Medium	Medium
Default Service Account Uses Editor Role	High	High
Security Command Center Unavailable	Informational	Low

Conclusion

The investigation successfully validated cloud-native monitoring and logging capabilities within Google Cloud Platform. Audit logging, log analysis, detection engineering, alerting, and IAM security reviews were implemented using native GCP services. The project demonstrated practical cloud security engineering skills including security monitoring, audit analysis, access control assessment, and security alert configuration.

The monitoring architecture provided visibility into cloud activities, established automated detection capabilities, and identified IAM security risks requiring remediation. The project reflects operational responsibilities commonly performed by Cloud Security Engineers and demonstrates foundational cloud security competencies applicable to enterprise environments.